🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 4 · STANDARD 6.AT.4

Determine the Whole Given the Part and Percent

Lesson 4-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve problems involving discounts, markups, tax, and tips using percents.
Language Objective: I can explain my solution using the words discount, markup, tax, tip, and percent.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Percent Problem Solving | Discount | Markup | Tax | Tip | Percent

👆 *Tap any word to see what it means and a picture.*

1. ______ — Use a percent to find a part, a whole, or a percent change in a real situation.
2. An amount taken off the original price is a ______.
3. The percent a store adds to its cost to set a higher selling price is a ______.
4. The percent added to a purchase that goes to the government is ______.
5. The percent of a bill you add for good service is a ______.
6. A number out of 100 is a ______.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

JUSTIFY

Is your friend correct? Show your work to prove it?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Percent Problem Solving** — Use a percent to find a part, a whole, or a percent change in a real situation.
  Resolución de problemas con porcentajes — Usar un porcentaje para hallar una parte, un total o un cambio porcentual en una situación real.

- [ ] **Discount** — Money taken off the first price to make it cheaper.
  Descuento — Dinero que se quita del precio original para que sea más barato.

- [ ] **Markup** — The extra amount a store adds to what it paid, found by taking a percent of that cost. Cost + markup = selling price.
  Recargo (margen) — La cantidad extra que una tienda suma a lo que pagó, hallada tomando un porcentaje de ese costo. Costo + recargo = precio de venta.

- [ ] **Tax** — Extra money added to the price of a purchase for the government, found by taking a percent of the price. Price + tax = total.
  Impuesto — Dinero extra que se suma al precio de una compra para el gobierno, hallado tomando un porcentaje del precio. Precio + impuesto = total.

- [ ] **Tip** — Extra money you add for good service, found by taking a percent of the bill. Bill + tip = total.
  Propina — Dinero extra que agregas por un buen servicio, hallado tomando un porcentaje de la cuenta. Cuenta + propina = total.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

First I find the discount: 30% of $50 = $____. The sale price is $50 − $____ = $____. Then I add 6% tax: $____ × 0.06 = $____. The total is $____ + $____ = $____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A discount lowers the price, while tax, tip, and markup raise it.

Evidence: Students sort discount as lowering price and tax/tip/markup as raising it, and recognize all are a percent of a price.

Reasoning: This evidence connects the definition of percent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **First I find the discount: 30% of $50 = $____. The sale price is $50 − $____ = $____. Then I add 6% tax: $____ × 0.06 = $____. The total is $____ + $____ = $____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Percent Problem Solving
2. **Notes 2:** Discount
3. **Notes 3:** Markup
4. **Notes 4:** Tax
5. **Notes 5:** Tip
6. **Notes 6:** Percent
7. **Watch for this mistake:** *A common mistake in Use Percent to Solve Problems is combining the discount and tax into one single percent change instead of applying them one after another. For a \$40 item with a 25% discount and 8% sales tax, a student might compute 40 × (1 − 0.25 + 0.08) = 40 × 0.83 = \$33.20, treating the tax as if it applies to the original price. But tax is charged on the discounted price, not the original price: first take 25% off \$40 to get \$40 × 0.75 = \$30, then add 8% tax on that \$30: \$30 × 1.08 = \$32.40. The correct final price is \$32.40, not \$33.20 — always compute the discount first, then apply tax to the new lower price.*
8. **Try It 1:** A. \$27 — *Tax: 0.08 x 25 = 2. Total: 25 + 2 = \$27.*
9. **Try It 2:** A. \$6 — *Tip: 0.15 x 40 = \$6.*